# ARANYA AI

Distributed Acoustic Intelligence for Forest Protection

**Complete Smart India Hackathon (SIH) Technical & Product Guide**
Problem • Architecture • Hardware • AI • Mathematics • Research • Competitors • Validation • SIH Strategy

## Executive summary

Aranya AI is proposed as an edge-AI acoustic early-warning network for remote forests. Low-power acoustic nodes listen locally, classify important sound events, and transmit compact event metadata rather than continuous raw audio. A central dashboard shows event type, confidence, sensor, time, battery and location/zone so forest personnel can investigate. The initial SIH prototype should focus on a small number of classes—chainsaw/tree-cutting-like sounds, vehicles, selected wildlife vocalizations, fire-related acoustic/environmental events, and background—and demonstrate a 2–3 node network.

**Important positioning:** acoustic forest monitoring is an established research and commercial field. Aranya should therefore not claim that acoustic monitoring is new. Its proposed differentiation is an India-focused, low-cost, modular, edge-first, multi-event system combining local AI, temporal confidence, distributed sensing, low-bandwidth alerts, solar operation, privacy-aware event transmission and an actionable ranger dashboard.

**Research basis:** autonomous real-time acoustic networks have already been demonstrated in tropical forests, including solar-powered Raspberry Pi devices; modern work also demonstrates on-device bioacoustic AI and Indian passive-acoustic monitoring. These precedents validate feasibility while making differentiation essential.

## 1. THE PROBLEM

### The core problem

India's forests are large, remote and difficult to monitor continuously. Patrol teams, camera traps, drones, satellite imagery and human reports are valuable, but each has coverage, endurance, visibility, connectivity or response-time limitations. A camera only observes its field of view; a patrol cannot be everywhere; a drone cannot remain continuously airborne; satellite imagery is not a replacement for local, immediate event detection.

### The specific gap

Aranya should define the problem as detection latency: **How can forest authorities receive an early warning when a potentially important event begins in a remote area, including events outside a camera's field of view?** Sound becomes an additional sensing layer.

### Who benefits

Forest departments and rangers; wildlife and biodiversity; communities near forests; conservation researchers; emergency-response teams dealing with fires or other hazards.

### Use cases

Chainsaw/tree-cutting-like acoustic activity; vehicle/engine intrusion; selected wildlife vocalizations; fire-related acoustic/environmental signals; unusual mechanical or human activity. The AI provides an alert for human verification—it does not declare that a crime has occurred.

## 2. WHAT ARANYA AI IS

## One-line definition

**Aranya AI is a distributed, solar-powered, edge-AI acoustic early-warning network for remote forest protection.**

## Core workflow

Forest sound → MEMS microphone → signal processing → log-Mel spectrogram/features → lightweight AI → confidence/temporal filtering → sensor/network transmission → ranger dashboard → human investigation.

## The key idea

Instead of asking only “Can we see the problem?”, Aranya asks “Can the forest tell us that something unusual may be happening?”

## What it is not

It is not an autonomous enforcement system, a guaranteed fire detector, a system that can identify a person from sound, or a replacement for rangers. These distinctions should be explicit in the SIH pitch.

# 3. SYSTEM ARCHITECTURE

## Sensor node

Microphone captures sound. An edge computer or microcontroller preprocesses it and runs the model. A power-management system controls battery/solar input. A communications module transmits event metadata. GPS/time synchronization and environmental sensors can provide context.

## Edge processing

The device processes audio locally. This reduces bandwidth, latency and privacy exposure. Continuous raw audio does not need to be transmitted.

## Network

LoRa is suitable for low-bandwidth alert metadata; cellular can be used where available for richer data. LoRa should send alerts, not continuous WAV audio.

## Dashboard

Map-based view with green/amber/red sensors; event type; confidence; timestamp; duration; sensor ID; battery; temperature; spectrogram/evidence; acknowledgement and incident history.

## Text architecture

FOREST SOUND → MEMS MICROPHONE → SIGNAL PROCESSING → EDGE AI → CONFIDENCE/TEMPORAL FILTER → LoRa/4G → RANGER DASHBOARD → HUMAN RESPONSE

# 4. HARDWARE

## Prototype recommendation

Use a Raspberry Pi for the first SIH build because it simplifies Python, audio processing and model deployment. A production direction can move to ESP32/STM32/TinyML-class hardware for lower power.

## Suggested parts

Raspberry Pi 4/5; INMP441 I²S MEMS microphone; LoRa 865 MHz module; 5–10 W solar panel; Li-ion/LiFePO4 battery; solar charge controller; 5 V/3.3 V regulator; microSD; GPS module; temperature/humidity sensor; IP65/IP67 enclosure; wiring/PCB/connectors.

## Approximate prototype budget

Raspberry Pi ■8,000–■14,000+; microphone ■150–■300; LoRa ■450–■2,500; solar ■500–■1,000; battery ■500–■1,500; controller/regulator ■100–■500; enclosure ■500–■1,500; GPS ■300–■800; storage and integration ■1,000–■2,000. A realistic 2–3 node prototype is roughly ■20,000–■35,000; a polished 3–5 node build may be ■40,000–■60,000 depending on component choices.

## Power design

Energy balance matters. $E_{daily} = P_{avg} \times 24$. Solar energy ≈ $P_{panel}$ × peak-sun-hours × efficiency. For example, 1.5 W average consumption requires 36 Wh/day. A 10 W panel with 4 effective sun-hours and 70% system efficiency provides about 28 Wh/day, so the design must reduce average power or increase generation/storage.

## Environmental design

The device needs rain, humidity, dust, heat, insects and physical protection. Use an appropriately sealed enclosure and an acoustic membrane/wind-noise solution. Validate rather than assuming outdoor performance.

# 5. AUDIO SIGNAL PROCESSING

## Sampling

Choose a sampling rate based on the target frequency range. Human-audible wildlife/anthropogenic events can often be represented around 16–48 kHz, but the actual rate should be selected from the dataset and microphone characteristics.

## STFT

For a discrete signal $x[n]$, the short-time Fourier transform can be written $X(m,k)=\Sigma\, x[n+mH]w[n]e^{(-j2\pi kn/N)}$. It converts a time signal into time-frequency information.

## Spectrogram

Power can be represented as $|X(m,k)|^2$. A Mel filterbank compresses frequency resolution according to perceptual spacing; log-Mel spectrograms are convenient inputs for CNNs.

## Why spectrograms

Different events have distinctive time-frequency patterns. A CNN can learn these patterns more easily than if the model were forced to discover all signal structure directly from raw audio.

## Preprocessing

Possible steps include normalization, optional band filtering, silence/activity screening, resampling and fixed-duration windows. Avoid preprocessing that removes information needed by the target classes.

# 6. AI MODEL

## Baseline

Start with a lightweight CNN on log-Mel spectrograms. It is easy to explain, train and deploy.

## Transfer learning

A stronger route is to use a pretrained bioacoustic embedding/model and train a small Aranya classifier. This reduces data and compute requirements. Perch 2.0 and BirdNET are useful research directions.

## Output

For C classes, softmax gives $P(y=c|x)=e^{(z_c)}/\Sigma_j e^{(z_j)}$. Example: Chainsaw 0.94, Vehicle 0.03, Wildlife 0.01, Fire 0.01, Background 0.01.

### Decision rule

Do not treat a single high score as proof. Require confidence above a threshold for multiple consecutive windows and/or fuse nearby sensors. A useful concept is temporal confidence aggregation: $C\blacksquare=(1/N)\Sigma C_i$, with an alert if $C\blacksquare$ exceeds a defined threshold for N windows.

### Multi-sensor fusion

If several nodes independently detect the same event in a short period, combine their confidence scores using weighted fusion $C=\Sigma w_i C_i$, $\Sigma w_i=1$. Later research can use Bayesian or learned fusion.

### Localization

With multiple synchronized microphones, time-difference-of-arrival can estimate source position. $\Delta d\approx v\Delta t$, with $v\approx 343$ m/s under standard conditions. Forest reflections, wind, temperature, terrain and clock synchronization make this difficult; the SIH MVP should initially report the sensor/zone rather than promise precise localization.

## 7. DATASET

### Positive classes

Chainsaw/tree-cutting-like sounds; vehicle/engine sounds; selected wildlife vocalizations; fire-related/environmental events.

### Negative/background classes

Rain, wind, insects, birds, branches, thunder, generators, motorcycles, distant vehicles and other natural/anthropogenic sounds. Background diversity is essential.

### Data augmentation

Mix target signals with noise: $x_{aug}=x_{signal}+\alpha n$. Also vary amplitude, time shift, reverberation and, where justified, time/frequency masking.

### Indian relevance

Build a geographically diverse dataset where possible. Research in India's Western Ghats and recent Forest Owlet work demonstrate the value of passive acoustic monitoring in Indian forests. Project Dhvani has also evaluated acoustic indices in central Indian tropical dry forests.

### Data governance

Track source, location class, date, conditions, labeler and licensing. Keep train/validation/test splits geographically separated where possible to test generalization rather than memorization.

## 8. MATHEMATICS AND EVALUATION

### Precision

Precision=TP/(TP+FP). It answers: when the system raises this alert, how often is it correct?

### Recall

Recall=TP/(TP+FN). It answers: how many real target events did the system find?

### F1

F1=2(Precision×Recall)/(Precision+Recall). Useful when both false alarms and missed detections matter.

### Why accuracy is dangerous

A dataset dominated by background can make a useless classifier look excellent. For example, a model that always says “background” on 9,900 background samples and 100 target samples achieves 99% accuracy but detects nothing.

### Operational metrics

Report false alerts per sensor per day, detection latency, inference time, power consumption, packet-delivery rate, communication range and cost per node.

### Latency

$T_{latency}=T_{alert}-T_{event}$. Minimize this without creating excessive false alarms.

### Power

P=VI. Measure average, peak and sleep-mode power experimentally rather than relying only on component datasheets.

### Distance testing

Test detection at controlled distances such as 10, 25, 50, 75 and 100 m, then report actual performance. Do not claim a universal acoustic range.

### Robustness testing

Repeat with rain, wind, insects, overlapping sounds and different microphone orientations.

## 9. COMMUNICATION AND PRIVACY

### Metadata-first communication

Transmit sensor ID, timestamp, event class, confidence, duration, battery, environmental context and location/zone rather than continuous raw audio.

### Privacy

Local processing reduces unnecessary transmission of human speech. Raw audio should be retained/transmitted only under an appropriate deployment and governance policy.

### Cybersecurity

Use unique device identities, encrypted communication, secure credentials, signed firmware updates, audit logs and tamper-aware design.

### Connectivity fallback

Where cellular coverage is absent, LoRa gateways or other suitable low-bandwidth architectures can carry alerts. Satellite should be treated as a future option, not an MVP requirement.

## 10. COMPETITIVE LANDSCAPE

### AudioMoth

A low-cost, open acoustic recorder widely used for biodiversity research. Its strength is recording and low-power field deployment. Aranya's proposed difference is real-time edge event classification and actionable alerts rather than primarily data logging.

### Wildlife Acoustics Song Meter

Professional commercial acoustic recorders with rugged field deployment. Strong recording hardware; Aranya targets low-cost event intelligence and alerting.

### Rainforest Connection Guardian

A major precedent for solar-powered, connected acoustic forest monitoring and illegal-activity detection. Aranya must acknowledge this directly. Its differentiation must be India-focused deployment, modular low-cost architecture, local edge intelligence, multi-event classification and experimentally demonstrated operating metrics.

### BirdNET

Strong biodiversity/species acoustic identification. Aranya can potentially use similar embeddings or transfer-learning ideas but focuses on forest threat/event detection and operational alerts.

### HARK/other edge systems

Demonstrate that on-device wildlife acoustic AI is feasible. Again, Aranya's innovation must be the end-to-end application and deployment design.

### Competitive claim to use

Do not say “world's first” or “no existing solution.” Say: **“Aranya integrates established acoustic AI, edge computing and low-power networking into a modular, India-focused multi-event early-warning platform, optimized for local soundscapes, low bandwidth and actionable ranger response.”**

## 11. RESEARCH PAPERS AND TECHNICAL FOUNDATIONS

### SAFE Acoustics (Sethi et al., 2020)

Open-source, real-time eco-acoustic monitoring network in tropical Borneo; deployed Raspberry Pi-based solar-powered devices and remote data infrastructure. Strong precedent for the end-to-end architecture.

### acoupi (Vuilliomenet et al., 2026)

Open-source framework for deploying bioacoustic AI models on edge devices, including Raspberry Pi; directly relevant to local inference and transmitting only relevant information.

### Western Ghats passive acoustic monitoring (2023)

Demonstrates passive acoustic monitoring for multi-faunal restoration assessment in southern India's Western Ghats.

### Forest Owlet acoustic monitoring (2024)

Shows automated acoustic detection methods for an endangered Indian bird and reports effective detection distance under its study conditions.

### Project Dhvani / central India (2024)

Evaluates acoustic indices in tropical dry forests of central India, supporting the relevance of soundscape monitoring to Indian conservation.

### Full-Stack Bioacoustics (2022)

Workshop report describing the need to integrate field hardware, AI and action into complete bioacoustic systems.

### Noise-robust sound event detection

Research on robust bioacoustic detection highlights environmental noise and context adaptation as central challenges.

### ForestIR (2026)

Physics-informed forest sound simulation for microphone-array bioacoustic remote sensing; relevant to future localization, synthetic data and array design.

### Key lesson from literature

The scientific premise is credible, but acoustic monitoring is not new. Your innovation is in system integration, deployment economics, Indian soundscapes, multi-event detection, low-bandwidth operation and actionable response.

## 12. SIH MVP

### Build only what matters

Three acoustic nodes + one gateway + one dashboard is better than a large unreliable system.

### Node

Raspberry Pi + MEMS microphone + LoRa + battery/solar subsystem + enclosure.

### Gateway

LoRa receiver connected to a laptop/Raspberry Pi/cloud dashboard.

### AI

5 classes: chainsaw-like, vehicle, wildlife, fire-related/background or a smaller subset if data quality requires it.

### Demo

Play a chainsaw clip → classification → confidence → sensor turns red on map. Play vehicle → different alert. Play wildlife → conservation event. Play background/rain → no high-priority alert.

### Proof

Show spectrogram, model output, latency, false-alert rate and battery status. Judges should see the complete chain, not only a dashboard mock-up.

## 13. DEVELOPMENT ROADMAP

### Phase 1

One Raspberry Pi + microphone + model + dashboard. Establish classification.

### Phase 2

Add LoRa and a second node.

### Phase 3

Three-node network and map-based alerts.

### Phase 4

Solar, battery, enclosure and power measurements.

### Phase 5

Temporal confidence, multi-node fusion and optional localization research.

### Phase 6

Field-style validation with varied Indian forest-like soundscapes and environmental noise.

# 14. RISKS AND HOW TO ANSWER JUDGES

## False positives

Use diverse negative/background data, temporal confirmation and multi-sensor fusion. Track false alerts/day.

## False negatives

Report recall and detection range; don't claim perfect detection.

## Dataset bias

Use geographic and environmental holdouts; collect more local data.

## Connectivity

Transmit tiny event packets; use LoRa/cellular depending on site.

## Power

Duty-cycle computation and communications; measure real energy consumption.

## Existing competitors

Acknowledge them. Differentiate on Indian deployment, multi-event threat/conservation classes, edge-first operation, low bandwidth and actionable response.

## Safety of deployment

Prototype only in controlled environments. Aranya provides decision support; humans verify and respond.

## Fire claims

Do not promise that acoustic monitoring alone detects every fire or detects fire before ignition. Treat fire as an additional early-warning signal and validate it separately.

# 15. SIH JUDGING STRATEGY

## Novelty

Emphasize integration and India-specific deployment rather than claiming acoustic monitoring is new.

## Complexity

Signal processing + AI + embedded computing + IoT + dashboard + power engineering.

## Feasibility

Small, demonstrable MVP with measurable metrics.

## Practicality

Low-bandwidth event packets, edge inference and human-in-the-loop alerts.

## Sustainability

Solar power and reduced data transmission.

## Scale

A few nodes can become a distributed forest network; architecture can extend from one protected area to multiple landscapes.

### User experience

Rangers receive concise, severity-ranked events instead of raw data.

### Future potential

Additional species, anthropogenic sounds, acoustic indices, localization, model updates and integration with existing forest control rooms.

## 16. WHAT NOT TO CLAIM

### Avoid

“World's first acoustic forest monitoring system.”

### Avoid

“Aranya detects poachers.”

### Avoid

“Aranya detects fires before they start.”

### Avoid

“99% accuracy” before measured validation.

### Use instead

“Detects acoustic events associated with potentially unauthorized or hazardous activity and alerts human authorities for verification.”

### Use instead

“Provides an additional acoustic/environmental early-warning layer.”

### Use instead

“Target performance will be validated using precision, recall, F1, false alerts/day and detection latency.”

## 17. RECOMMENDED SIH PITCH

### Opening

**A forest can be hundreds of square kilometres. A ranger cannot be everywhere.**

### Problem

Remote forest incidents can occur outside camera fields of view and far from patrol teams. The challenge is not merely surveillance—it is reducing the time between an event beginning and authorities becoming aware of it.

### Insight

Cameras provide visual coverage. Aranya adds acoustic coverage.

### Solution

Solar-powered sensor nodes listen locally, edge AI classifies important events, and only compact alerts are transmitted to a ranger dashboard.

### Innovation

Multi-event acoustic intelligence + edge AI + temporal confidence + distributed sensing + low-bandwidth communication + India-focused deployment.

### Impact

Potentially extend monitoring reach, reduce detection latency and give forest personnel a data-driven indication of where to investigate.

### Closing

**ARANYA AI — Hear the forest. Detect the threat. Protect it before it's too late.**

## 18. FINAL TEAM CHECKLIST

### Technical

Collect/curate labelled data; build baseline CNN; establish train/validation/test split; measure precision/recall/F1; measure false alerts/day; deploy inference on edge hardware.

### Hardware

Build microphone node; LoRa link; power system; enclosure; battery monitoring.

### Software

Dashboard; event database; map; alert logic; device health monitoring.

### Research

Read the cited papers; document what already exists; explicitly state your novelty.

### SIH

Define one exact problem; show a working prototype; quantify impact; explain deployment cost; present limitations; avoid unsupported statistics; have a live demo and backup video.

### Team roles

AI/data; embedded/hardware; networking/IoT; dashboard/backend; wildlife/domain research; pitch/business/deployment.

### Golden rule

A smaller system that works reliably is stronger than a giant architecture that exists only on slides.

# SELECTED WEB SOURCES FOR THE GUIDE

SAFE Acoustics — open-source real-time eco-acoustic network in tropical Borneo. ■cite■turn0search0■turn0search1■

acoupi — edge deployment of bioacoustic AI models on low-cost hardware. ■cite■turn0search7■

Western Ghats passive acoustic monitoring study. ■cite■turn0search10■

Forest Owlet acoustic monitoring in India. ■cite■turn0search3■

Project Dhvani / acoustic indices in central India. ■cite■turn0search6■

2026 soundscape management system using solar IoT acoustic devices and dashboards. ■cite■turn0search2■turn0search11■

Full-Stack Bioacoustics workshop report. ■cite■turn0academia15■

ForestIR physics-informed forest acoustic simulation. ■cite■turn0academia12■

Note: citations in this PDF are represented as source names and URLs should be checked before submission. The PDF is a planning/technical guide, not a guarantee of SIH acceptance or field performance.